

Resume

As of July 2026

Itaru Ota

Based in Tokyo, Japan

- [GitHub](#)
- [Portfolio](#)

Focus & Value

- Translate business requirements and product problems into designs spanning frontend, APIs, data models, and cloud infrastructure
- Design and implement backend services, APIs, and data models with changeability and operability in mind
- Establish reproducible development and release processes with Terraform, CI/CD, and automated testing
- Measure performance indicators to inform architecture migration and performance improvement decisions
- Clarify requirements and design decisions, and drive delivery through stakeholder alignment

Technical Expertise

- **Application design & development:** Python, TypeScript, Java, Rust, Django, gRPC, NestJS, Spring Boot, React, Next.js
- **Cloud & platform:** AWS, Azure, Google Cloud, Terraform, Docker, GitHub Actions
- **Data & AI:** PostgreSQL, MySQL, Redis, Prisma, Azure OpenAI Service, YOLO, MeCab, spaCy, OCR
- **Architecture & quality:** DDD, API and data-model design, CI/CD, automated testing with pytest and Jest

Work Experience

3-shake Inc.

Full-time Full-stack Engineer

Jan 2025 – Present

B2B SaaS Product Development

Jan 2025 – Present

- Delivered product features for a B2B SaaS product across frontend, backend, and cloud infrastructure
- Designed and implemented applications, APIs, and data models using Next.js, Django, and gRPC
- Improved AWS and Azure infrastructure management with Terraform, CI/CD, and automated testing
- Designed and implemented a gRPC API performance benchmark harness for a Python-to-Rust migration
- Compared throughput, latency, startup time, and memory usage, confirming performance improvements for the Rust implementation

Full-stack Development of a Machine Learning Development Support Application

Apr 2025 - Jun 2025

- Developed an application that supports machine learning model creation and improvement using Google Cloud, Next.js, and Django
- Designed and implemented both frontend and backend components

IT Technical Training Design and Delivery

Mar 2025

- Created training materials on Linux, Git, Docker, and GitHub Actions
- Served as an instructor and mentor for a five-day training program

Daiwa Institute of Research Ltd.

Full-time Application Engineer

Apr 2023 – Dec 2024

RAG System Development

Sep 2024 – Dec 2024

- Built and evaluated a RAG system using Azure OpenAI Service and Azure App Service

Core System API Modernization

Apr 2024 – Jun 2024

- Conducted basic and detailed design for long-running core system APIs
- Designed and implemented DDD-informed entity and repository structures
- Developed business-logic APIs using Java Spring MVC and maintained YAML API specifications
- Developed and tested in Jenkins, Subversion, and Oracle environments

CTI System Implementation

Oct 2023 – Aug 2024

- Gathered requirements for an incident-reporting phone system
- Designed and implemented contact- and time-based call routing with Amazon Connect
- Supported the transition from fixed-line-based reporting workflows to a CTI system

Incident Reporting System Modernization

Aug 2023 – Aug 2024

- Worked from requirements through release and operations, supporting ongoing improvements to a system that operates around the clock
- Implemented frontend changes with React, Next.js, and AWS Amplify, and email delivery with API Gateway, Lambda, S3, and SES
- Supported the systemization of reporting workflows previously managed through email and spreadsheets

ROE Forecasting Model Development

Oct 2023 – Jan 2024

- Analyzed economic data using MeCab-based preprocessing, pandas, NumPy, and Matplotlib
- Evaluated word2vec, BERT, and Amazon Bedrock Titan models
- Built a processing environment using AWS SageMaker and G4 instances

SOAT Inc.

Part-time Software Engineer

Oct 2021 – Mar 2023

Health Check Report PDF Structuring PoC

Jan 2023 – Mar 2023

- Implemented PDF image preprocessing with Python and OpenCV
- Implemented text extraction from health check reports with Tesseract OCR and data persistence in MySQL

Foreign Object Detection Model Improvement for Food Processing Lines

Apr 2022 – Dec 2022

- Improved training-data quality through image annotation with LabelImg and data preprocessing
- Compared YOLO models and improved generalization with OpenCV and data augmentation
- Built an evaluation framework for precision, recall, and F1 score with PyTorch

Document Difference Detection Application Development

Oct 2021 – Mar 2022

- Developed text extraction from PDF documents using Tesseract OCR
- Optimized image preprocessing with OpenCV to improve OCR accuracy
- Built a reproducible runtime environment with Docker and SQLite

Personal Projects

[momen.ai](#) - Conflict Cause Recommendation Web App

Feb 2025 – Mar 2025

- Built AI dialogue system using Next.js (App Router) and Vertex AI
- Implemented serverless architecture using Google Cloud Run and Artifact Registry
- Built an extensible and maintainable system through schema-driven design

Geolocation Guessing Board Game Web App

Aug 2023 – Oct 2024

- Full-stack development in TypeScript (React / Next.js + Node.js / NestJS)
- Adopted DDD layered architecture for scalability and maintainability
- Database design and type-safe implementation using Prisma, MySQL, and Redis
- Automated testing and builds through CI/CD pipeline with GitHub Actions

NLP OSS Development at Graduate Laboratory

Jun 2021 – Aug 2021

- Developed the Japanese text analysis library [limco](#)
- Built high-accuracy Japanese NLP pipeline using spaCy / GiNZA
- Implemented various linguistic metrics based on text structure and content
- Published the library on GitHub and PyPI

Community Activities & Presentations

2026 Japan AWS Jr. Champions

Jun 2026 - Present

- Selected for an AWS recognition program for early-career engineers at AWS Partner organizations
- One of 114 engineers selected nationwide for technical capability, organizational and community contributions, and public knowledge sharing through writing and presentations
- Published technical articles on Amazon ECS networking and Terraform-based Infrastructure as Code for AWS resources

Presentations

- [Meguro LT #37: "Reading the Why Behind Commits"](#), Jun 2026
- [3-shake SRE Tech Talk LT](#), Sep 2025
- [JAWS-UG Container Branch LT](#), Aug 2025
- [Qiita Event: "Bash - Let's Talk About Recent AI Use Cases"](#), May 2025

Certifications

- [AWS Certifications](#), 2023–2026
- Google Cloud (Associate Cloud Engineer, Generative AI Leader), 2025
- Applied Information Technology Engineer Examination, 2023
- Fundamental Information Technology Engineer Examination, 2023
- TOEIC Score 865, 2023

Education

- M.Eng. in Information Science, Nara Institute of Science and Technology (NAIST), Mar 2023
- B.A. in Economics, Keio University, Mar 2021

Graduate Research

Improving the Experience of Listening Broadcast Programs Using Social Media Data

Apr 2021 – Mar 2023

- Collected social media posts related to broadcast programs and analyzed post frequency and semantic similarity to identify highlight segments
- Designed and implemented a highlight-based listening experience by aligning the analysis results with audio content
- Conducted a user evaluation that compared the proposed experience with conventional full-length playback to examine accessibility and efficient listening of broadcast archives
- Presented research at DEIM 2022 and DEIM 2023; received the [DEIM 2022 Student Presentation Award](#) and the [imedia2021 Data Broad Award](#)